

Predicting the Ionospheric Cusp Location from Solar Wind

An XGBoost Model Trained on 27 Years of DMSP Data

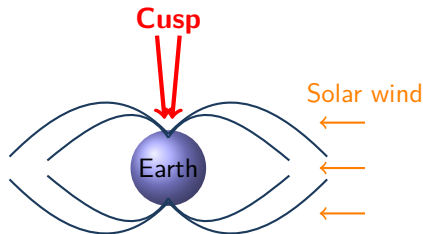
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What is the Cusp?

- ▶ Narrow funnel near each magnetic pole where solar wind plasma enters the magnetosphere along open field lines
- ▶ DMSP at ~ 840 km sees a $1\text{--}2^\circ$ MLAT band of low-energy ion precipitation (Newell criteria)
- ▶ Equatorward boundary tracks the dayside reconnection rate
- ▶ Practical impact: GPS scintillation, LEO drag, polar-route aviation radiation
- ▶ No operational model currently predicts cusp location from upstream solar wind



Where Existing Models Stand

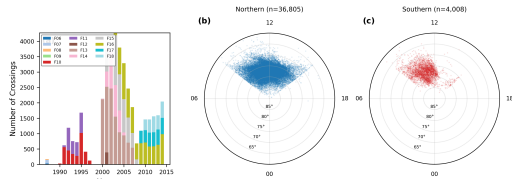
- ▶ **Newell et al. 2006**: 10 coupling functions tested. Best (E_{WAV}): $r \approx 0.56$ on individual crossings, 0.81 on binned data. 60-min averaging beats instantaneous.
- ▶ **Newell & Sotirelis 2007**: universal coupling function $d\Phi_{MP}/dt$. $r = 0.85$ binned.
- ▶ **Anderson & Bukowski 2024**: 41k DMSP crossings; tilt slope varies with MLT and IMF B_y sign.
- ▶ **Global MHD**: faithful but hours of CPU per run; not operational.

The gap: every published predictor uses 1–2 SW parameters and linear / low-order nonlinear fits. Crossing-to-crossing scatter is large ($r \sim 0.56$ – 0.58). Multi-parameter ML has not been applied.

Data — DMSP + OMNI

- ▶ **DMSP F06–F18**, 1987–2014 (27 yr ≈ 2.5 solar cycles)
- ▶ SSJ/4 + SSJ/5 spectrometers, 1-s cadence
- ▶ Anderson cusp criteria $\rightarrow$ **39,668 crossings** (clean, AE-filtered)
- ▶ Along-track boundaries: eq/pole MLAT + eq/mean MLT at the satellite's MLT
- ▶ *This is a 1D snapshot at one MLT, not the full 2D cusp*

OMNI L1: 1-min IMF + SW $\rightarrow$ **74 features** including 15/30/60-min mean/std/ Δ history



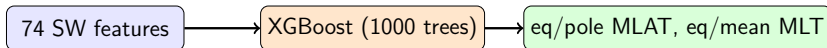
Input Features (74 Total)

Category	#	Key features
Geometry	3	dipole tilt, hemisphere, day-of-year
Raw IMF	3	B_x , B_y , B_z
Solar wind	3	v , n , P_{dyn}
Coupling	7	$d\Phi_{MP}/dt$, E_{KL} , vB_s , B_T , θ_c , $B_y \cdot \text{sgn}(\text{hemi})$, $\sin(\theta_c/2)$
15-min history	18	mean, std, Δ of 6 IMF/SW vars
30-min history	18	mean, std, Δ of 6 IMF/SW vars
60-min history	22	mean, std, Δ + integrals of $d\Phi_{MP}/dt$, vB_s
Total	74	

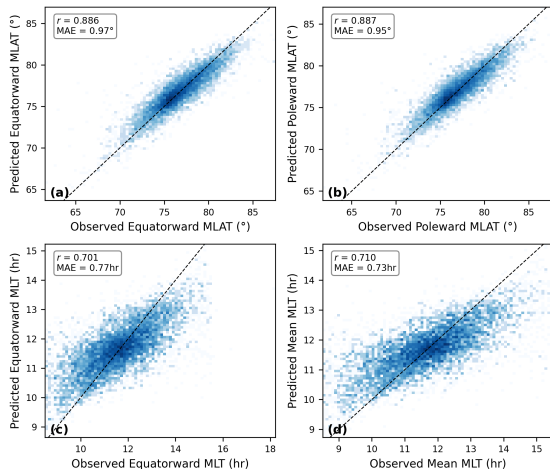
Key choice: include time-integrated history because the magnetosphere takes ~ 1 hr to reconfigure after an IMF change.

Method — XGBoost on Tabular SW → Cusp

- ▶ **XGBoost**: gradient-boosted decision trees. 1000 trees, depth 8, lr 0.02, subsample 0.8, colsample 0.7.
- ▶ Tree models beat NNs on tabular data (Grinsztajn et al. 2022); we confirmed against MLP, ResMLP, TabTransformer.
- ▶ Targets (multi-output): eq MLAT, pole MLAT, eq MLT, mean MLT.
- ▶ Interpretability: gain importance + SHAP + partial dependence.



Headline Result — Temporal Holdout



Train pre-2008 / test 2008–2014

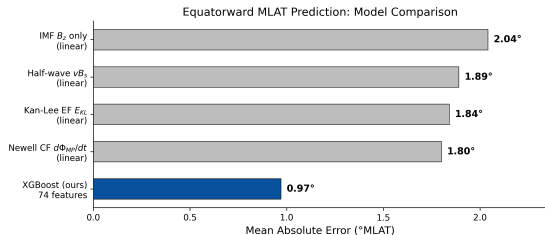
Target	MAE	R^2	r
eq MLAT (°)	1.11	0.79	0.89
pole MLAT (°)	1.09	0.79	0.89
eq MLT (hr)	0.77	0.48	0.70
mean MLT (hr)	0.73	0.49	0.71

$1.11^\circ \approx 122 \text{ km on the dial}$

$\approx 60\%$ of crossings within 1° , 90% within 2°

Random-split (paper-1 backup): $MAE = 0.97^\circ$.

Generalization & Baseline Comparison



Leave-one-year-out (23 years):

- ▶ $MAE = 1.26 \pm 0.19^\circ$
- ▶ Robust across solar cycle phases

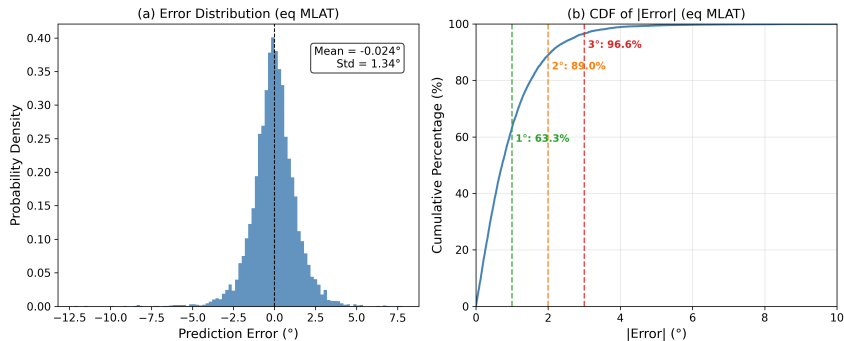
vs. baselines (temporal holdout):

Method	MAE	r
Linear (B_z)	2.04°	0.39
Linear (Newell CF)	1.85°	0.58
Nonlinear Newell	1.80°	0.59
XGBoost	1.11°	0.89

40% MAE reduction over best linear baseline

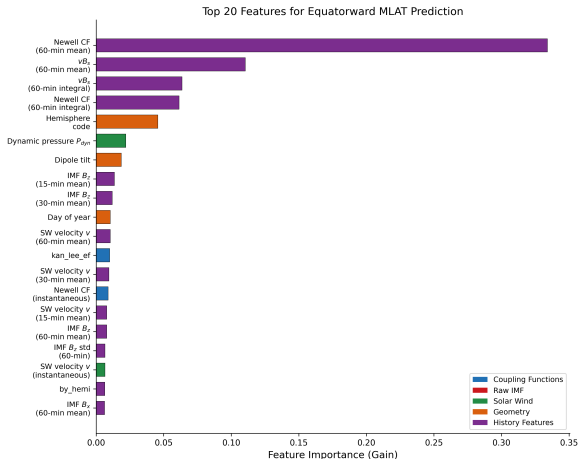
$p < 0.001$

Error Distribution



Residuals: mean $< 0.03^\circ$ (near-zero bias), std $\approx 1.34^\circ$. Multi-crossing within-group spread $\sim 1.03^\circ \Rightarrow$ model MAE approaches the natural cusp variability floor.

What the Model Learned



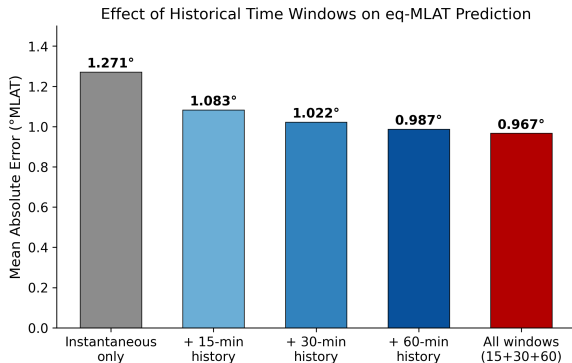
Top predictors:

- ▶ **#1** ($\sim 31\%$): 60-min mean $d\Phi_{MP}/dt$
- ▶ **#2** ($\sim 11\%$): 60-min mean vB_s
- ▶ **#3-4** ($\sim 6\%$ each): 60-min integrals of $d\Phi_{MP}/dt$, vB_s
- ▶ Hemisphere ($\sim 5\%$), dipole tilt ($\sim 2\%$)

60-min $\gg$ 30-min $>$ 15-min $\gg$ instantaneous

Model rediscovered the established physics on its own.

How Much History Matters?



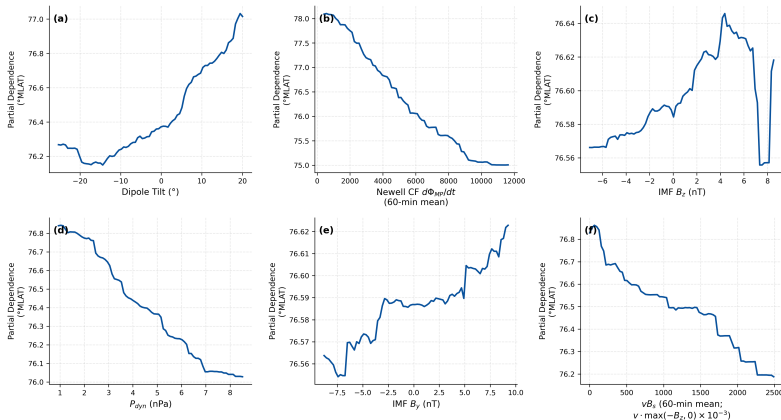
Feature set	MAE
Instantaneous only	1.27°
+ 15-min history	1.08°
+ 30-min history	1.02°
+ 60-min history	0.99°
All windows combined	0.97°

60-min history gives the single largest gain (-0.20°).

Matches the ~ 1 -hr magnetospheric reconfiguration timescale.

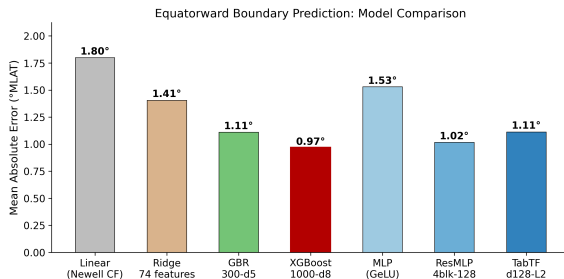
Physical Relationships — Partial Dependence

Partial Dependence of Equatorward Boundary MLAT



(a) Dipole tilt slope $-0.05^\circ/^\circ$ (Newell 1989). (b) Newell CF nonlinear saturation. (c) B_z asymmetry. (d) P_{dyn} . (e) B_y . (f) vB_s . Each panel recovers a known empirical relationship without being told the formula.

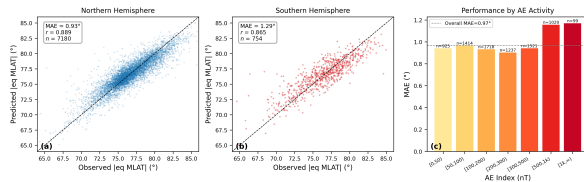
XGBoost vs. Neural Networks



Model	MAE	Train
Linear (Newell)	1.83°	0.1 s
GBR-300	1.11°	–
XGBoost	0.97°	80 s
MLP-GeLU	1.54°	11 s
ResMLP	1.01°	36 s
TabTransformer	1.11°	136 s

Tree models beat NNs on this tabular task (Grinsztajn et al. 2022).

Robustness: Hemisphere & Activity



Hemisphere (random split):
N ($n \approx 7180$, 90%): MAE 0.93°
S ($n \approx 754$, 10%): MAE 1.29°

Geomagnetic activity (AE):

AE level	MAE
Quiet (<100)	0.96°
Moderate (100–300)	0.92°
Active (300–500)	0.94°
Storm (≥ 500)	1.16°

Storm: +25% MAE; still well below the 1.83° linear baseline.

Limitations & Operational Path

Scope of the claim:

- ▶ Along-track 1D boundary at the satellite's MLT
- ▶ Not a full 2D cusp map (work in progress)
- ▶ AE-filtered crossings (Anderson 2024 criterion)
- ▶ 1987–2014 only; rare extremes under-represented

Real-time path: all 74 inputs are derivable from upstream L1 monitors (ACE/DSCOVR). Requires validation on real-time L1 data.

Take-home:

- ▶ First ML cusp predictor: **1.11°** temporal holdout, $1.26 \pm 0.19^\circ$ LOYO
- ▶ 40% MAE reduction over best linear baseline
- ▶ Model rediscovered known physics: 60-min Newell CF dominance, dipole tilt slope, CF saturation
- ▶ Path to real-time forecasting from L1 SW monitors

Data & code: Zenodo DOI
10.5281/zenodo.19340792
GitHub: Yiwenzhu77/cuspML